

Shane Thakkar

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SUMMARY

Analytics & AI professional with 2.5 years of hands-on experience in Python, SQL, and machine learning across sports, finance, and telecom. Built automations saving 10+ hours per week, delivered ML models reaching 90% accuracy, and developed venue clustering that guided multimillion-dollar LA28 investments. Independently ship statistically rigorous full-stack data products, including a public NFL player grading platform with 20,000+ player-season grades, Bayesian shrinkage, and audit-defensible methodology.

EDUCATION

B.S. Business Analytics and Artificial Intelligence – Data Science Track

University of Texas at Dallas • Richardson, TX • May 2026

SKILLS

Programming & Analytics: Python, Pandas, NumPy, SciPy, TypeScript, SQL, R, Git, Bash/UNIX, Jupyter

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost, Optuna, LangChain, Transformers

Statistical Methods: Time-Series Forecasting, Clustering, Regression Analysis, Econometrics

Big Data & Cloud: Databricks, Hadoop, Spark, Hive, Snowflake, Azure, SQL Server

Visualization & Tools: Tableau, Power BI, Matplotlib, Plotly, Excel, Stata, Alteryx, Power Automate

EXPERIENCE

Intern - Business Intelligence

Legends Global

July 2025 - December 2025, Frisco, TX

- Analyzed 1M+ POS records across 8+ professional sports venues using Python and Pandas, cleaning and standardizing data to perform category-level performance analysis that identified high and low-performing products and drove inventory, pricing, and seasonal planning decisions.
- Applied Market Basket Analysis and Pareto principles in Python to evaluate FC Dallas concessions profitability, discovering high-frequency product pairings for bundling strategies and identifying 30% of SKUs as tail products generating <15% of revenue.
- Consolidated 2 years of POS data from 3 systems for Del Mar Racetrack, building reports and SWOT analyses that identified key revenue drivers and performance trends, presenting findings to stakeholders to inform operational strategy.
- Modeled F&B performance for ~15 LA28 venues in Python by normalizing historical per-cap and attendance data, then clustering venues into priority tiers that informed executive investment and risk decisions.
- Automated weekly SQL Server reconciliation between game-day flash reports and POS transaction data across venues, flagging discrepancies and data gaps in audit logs and eliminating ~10 hours per week of manual validation work.
- Pioneered transparent reporting practices by adding methodology sections to all analyses, documenting data sources, assumptions, and formulas for stakeholder validation; later adopted as the team standard.

Intern - Project Management and Data Analytics

Supreme Lending

June 2024 - November 2024, Dallas, TX

- Built Excel-based project management system coordinating 100+ test cases across departments during loan origination software upgrade, tracking completion KPIs through automated dashboards that enabled defect-free production rollout.
- Built Power Automate workflow to automate Azure DevOps user story creation, reducing manual request intake time and enabling automated stakeholder tracking and notifications.
- Led team of 14 interns in 30-minute executive pitch proposing university mortgage lending course, presenting Excel pivot-table analysis of 300+ student surveys showing ~60% interest.

Intern - Business Analyst

TruGen

May 2023 - May 2024, Frisco, TX

- Cleaned and structured wireless network performance data across 10+ regions using SQL, preparing datasets for machine learning models that achieved 90% classification accuracy in signal quality prediction.

RELEVANT PROJECTS

NFL Player Grading Platform

nfl-grades.shanethakkar.com

Built and deployed a full-stack NFL analytics data product generating 0–100 player grades by position and season across 12 positions and 2016–2025 data, producing 20,000+ player-season grades.

- Engineered a six-stage Python grading pipeline using pandas, SciPy, and SQLAlchemy, applying volume-weighted empirical Bayes shrinkage, qualified-cohort z-score normalization, magnitude-normalized composites, sigmoid calibration, and Kalman smoothing to stabilize grades.
- Evaluated 190+ candidate metrics against four criteria: year-over-year reliability, cross-sectional discrimination, independence, and predictive validity vs. Pro Bowl selections; shipped 52 components to production at a 27% acceptance rate with rejections logged for audit defensibility.
- Deployed a Next.js 15, React 19, and TypeScript web app on Vercel with Neon serverless Postgres, 25+ idempotent SQL migrations, mobile-responsive sortable leaderboards, 647 AI-generated player headshots via the Replicate API, and composite-indexed queries returning in under 50ms.

F1 Bayesian Driver Rankings

shanethakkar.com/articles/f1

Built a hierarchical Bayesian model decomposing driver skill from car performance across 12 seasons of Formula 1 race data from 2014–2025, covering 46 drivers and 123 team-seasons.

- Engineered the model in Python using PyMC and FastF1, applying a Student-T likelihood for heavy tails, partial pooling hyperpriors on driver and team-season variance, and a DNF-fault dummy to prevent penalizing drivers for mechanical failures outside their control.
- Validated convergence with $\hat{R} \approx 1.0$ across NUTS chains, temporal holdout RMSE of 3.114 vs. baseline 3.146, and posterior predictive checks reproducing the empirical residual distribution.

NFL Fourth-Down Decision Analysis

shanethakkar.com/articles/fourth-down

Analyzed 107,000+ NFL fourth-down decisions from 1999–2025 using Win Probability Added to grade 167 head coaches on decision quality, with three deployed interactive web apps and a published article.

- Defined original Decision Quality Score and Optimal Decision Rate metrics on a WPA-based framework, binning 107,000+ plays across a four-dimensional state lattice of field position, yards-to-go, score differential, and time remaining with exponential recency weighting and minimum-sample filtering.
- Trained three XGBoost regressors predicting expected WPA for go-for-it, punt, and field-goal decisions on 1999–2023 data with 2024–2025 held out for temporal validation.
- Deployed three interactive Streamlit applications: a decision calculator that scores any game-state input, an XGBoost-powered decision boundary heatmap visualizing optimal calls across the field, and a sortable explorer ranking 167 head coaches by Decision Quality Score and Optimal Decision Rate.

SEC 10-K Risk Factor Analyzer

Built a multi-agent AI system in Claude Code automating equity research on SEC 10-K filings: retrieving from EDGAR, classifying risk changes, scanning distress signals, and generating analyst-ready investor memos.

- Architected a two-agent workflow with explicit handoff protocols across six Python skills: a Filing Agent handling EDGAR retrieval, HTML parsing, and fuzzy-matching year-over-year diff; an Analyst Agent handling eight-category severity-scored classification, distress-language scanning, and investor memo synthesis.
- Demonstrated on Boeing's FY2024 and FY2025 10-K filings, identifying new operational and regulatory risks from 737 MAX manufacturing issues and FAA scrutiny, removal of pandemic-era recovery language, and a year-over-year jump in regulatory-investigation red flags.

CERTIFICATIONS

IBM AI Engineering Professional Certificate · Snowflake Hands-On Essentials: Data Warehousing Workshop · Professional Scrum Master™ I